

Department of Computer Science

Federal University Lafia

First Semester Examination

Session: 2016/2017

CSC111: Introduction to Computer Science

Date: April, 2017 Time: 2hrs.

Instruction: Attempt all questions. Each question carries 1 mark. Only Shade the correct option in the attached answer sheet. Clearly write your matriculation number and your department in the space provided on the answer sheet. The answer sheet should be carefully detached and submitted at the end of the examination.

1. A _____ is a collection of related Web pages
a. Internet b. Web browser ☒ c. Website d. Directory
2. As a 100L student, what would you use when creating and manipulating documents?
a. Document software ☒ b. Word processing software c. Document manipulation software d. System software.
3. A _____ is a stationary pointing device with a ball on its top or side and about the size of a Ping-Pong ball.
☒ a. mouse b. keyboard c. touchpad d. trackball
4. ExpressCard module is a storage device that fits in an expressCard slot and is developed by the
☒ a. ISA b. PCI c. PCMCIA d. IBM
5. _____ allows people to connect wirelessly to the Internet from home, work, school, and in many public locations
a. wireless internet application b. wireless interpreter ☒ c. Wireless Internet access points d. Application software
6. _____ is a collection of data organized in a manner that allows access, retrieval, and use of that data
a. transaction ☒ b. database c. information d. data
7. A _____ refers to the layout of the computers and devices in a communications network
a. network points b. network layout c. Network setup ☒ d. Network Topology
8. A _____ is a pressure-sensitive pointing device shaped like a pencil eraser that is positioned between keys on a keyboard
☒ a. trackball b. mouse c. pointing stick d. touchpad
9. The _____ holds data waiting to be processed and instructions waiting to be executed.
a. flash drives b. registers c. processor ☒ d. main memory
10. One of the following is NOT an item in the information processing cycle
a. Collection of data ☒ b. Detection of error c. Production of information d. processing
11. The resolution of an Ink-Jet printer can be measured by the number of _____ the printer can print.
☒ a. dots per inch b. pixels per inch c. pages per minute d. line per minute
12. _____ tells the computer what task to perform and how to perform them.
a. Boot-Drive b. Controller c. Processor ☒ d. Software
13. A _____ is a collection of computers and devices connected together via communications devices and transmission media
☒ a. network b. operating system c. hard disk d. Multimedia
14. Which of the following is a graphic file format?
a. exp b. fgi ☒ c. jpg d. bpm
15. _____ is the full meaning of PSTN
☒ a. Public switched telephone network b. public swift telecoms network c. public service telephone network
d. private switched telephone network
16. The program that allows web pages to be viewed is _____
a. Website ☒ b. Web browser c. Search engine d. World wide web
17. The _____ chip, which uses battery power, stores configuration information about the computer.
a. CMOS ☒ b. BIOS c. POST d. RAM
18. _____ is a network that connects computers and devices in a limited geographical area
a. WAN b. MAN ☒ c. LAN d. PAN

19. A _____ is a group of related fields in a database.
 a. character b. record c. primary key d. data type
20. A _____ is an input device that allows a user to control a pointer on the screen.
 a. mouse b. keyboard c. touchpad d. all of the above
21. To organize data in rows and columns and perform calculations, users need to have _____.
 a. Data software b. Spreadsheet software c. Business software d. typing software.
22. _____ is a number that uniquely identifies each computer or device connected to Internet
 a. Network address b. IP address c. System address d. Device address
23. One of these is NOT a strategy employed to achieving green computing.
 a. Recycling b. Proper disposal of replaced computers c. Minimizing the life of computers d. Regulating manufacturing processes
24. Which of the following is not an advantage of solid state drives over magnetic hard disk.
 a. faster access time b. faster transfer rate c. short life span d. less power consumption.
25. Turning on a computer that has been powered off completely is called
 a. hard boot b. soft boot c. cold boot d. power boot
26. _____ is text or graphic that displays a related Web page
 a. Hyperlink b. Domain name c. URL d. search engine
27. One of the following is not a basic type of output.
 a. text b. data c. graphics d. audio
28. All these are varieties of Application software EXCEPT;
 a. Packaged software b. Custom software c. Open source software d. Managed software.
29. _____ is the full meaning of URL
 a. Unified Resource Locator b. Uniform Resource Locator c. Unified Release Locator d. Uniform Release Locator
30. _____ is not a function of a DBMS
 a. to create a computerized database b. Sort and retrieve data c. Booting a system d. none of the above
31. The _____ is a case that contains electronic components of the computer used to process data.
 a. System unit b. arithmetic logic unit c. monitor d. printer
32. _____ device is a server connected with a network with a sole purpose of providing storage.
 a. NIC b. NAS c. RAID d. SSDs
33. A technology that sends resistance to a device in response to actions of the user is known as _____.
 a. interactive whiteboard b. CRT c. force feedback d. dance pad
34. ISP refers to _____
 a. Internet Site Provider b. Internet Sales Provider c. Inter network Sales Provider d. Internet Service Provider
35. Domain name refers to _____ version of IP address
 a. Graphic b. Similar c. Duplicate d. Text
36. A large expensive, powerful computer that can handle hundreds or thousands of connected users simultaneously is called:
 a. Mainframe b. Server c. Supercomputer d. Mega-computer
37. All the following are examples of commonly used audio output except one.
 a. Speakers b. Earbuds c. Monitor d. Headphones
38. All of these are challenges or disadvantages associated with using computers EXCEPT;
 a. Enhanced Privacy b. Labour force depletion c. Environmental impact d. Health risk
39. Which of the following is not a reason why user subscribes to cloud?
 a. Access file from any computer b. store large file instantaneously c. stores onsite backup d. provides datacenter function.
40. The following are functions of Application software EXCEPT?
 a. To make business activities more efficient. b. To assist with graphics and multimedia projects. c. To support home, personal and educational tasks. d. To fix computer hardware and network issues.
41. _____ saves any open documents and programs to RAM, turns off all unneeded functions, and then places the computer in a low-power state
 a. low power mode b. safe mode c. sleep mode d. hibernate

42. A _____ is the smallest unit of data the computer can process.
 a. character b. byte ☒ c. bit d. word
43. _____ is the full meaning of GPS.
 a. global point system b. global parameter system ☒ c. global positioning system d. global print system
44. A _____ is a small, flat, rectangular pointing device that is sensitive to pressure and motion.
 a. trackball b. mouse ☒ c. touchpad d. Keyboard
45. GUI standards for _____
☒ a. graphical user interface b. going user interface c. group user interface d. game user interface
46. _____ is a set of programs containing instructions that work together to coordinate all the activities among computer hardware resources.
☒ a. operating system b. RAM c. database d. antivirus
47. wi-fi is sometimes referred to as _____
 a. 802.67 b. 802.30 ☒ c. 802.11 d. 802.20
48. _____ describes an Internet network on which users access each other's hard disks and exchange files directly over the Internet.
 a. WAN ☒ b. P2P c. WLAN d. Token Ring
49. A(n) _____ device is any hardware component that allows users to enter data and instructions (programs, commands, and user responses) into a computer.
 a. output b. storage ☒ c. input d. communication
50. _____ Serves as the intermediary between the user, the application software, and the computer's hardware.
 a. Computer system b. Cloud computing ☒ c. Operating System d. hybrid software.
51. ARPANET refers to _____
 a. Advanced Reserve Agency network b. Advance Reverse Projects Agency network ☒ c. Advanced Research Projects Agency Network d. Advance Research Projects Agency network
52. _____ is an alternative to DVI that also supports HDMI.
 a. standard monitor port b. DisplayPort c. Digital Video Interface d. S-video port
53. _____ is the full meaning of DBMS
☒ a. database management system b. database moving system c. database mirroring system d. database manipulation system
54. All the following are output devices except one.
 a. Interactive whiteboards b. Data projectors ☒ c. Light Pen d. Earbuds
55. The processor also is called the _____.
 a. motherboard ☒ b. central processing unit (CPU) c. adapter card d. chip
56. _____ converts signals between analog and digital.
 a. converter b. reconverter ☒ c. modem d. deconverter
57. Output is _____ which is processed into useful information.
 a. text b. video c. graphics ☒ d. data
58. _____ is not a common data type
☒ a. memo b. object c. attachment d. variable
59. The number of byte a storage medium can hold is called _____.
 a. Density ☒ b. storage density c. Access time ☒ d. Capacity
60. Which of the following is not a type of flash memory storage?
 a. microSD ☒ b. memory stick digital c. secure digital d. xD picture card

2051600012

GENERAL STUDIES
BATCH B

FIRST TEST ON GST 113: NIGERIAN PEOPLE AND CULTURE

DEPARTMENT: MICROBIOLOGY

MATRIC NUMBER: 2051600012 TIME ALLOWED: 40 MINUTES

9
15

1. Culture is a learned disposition to behave in a consistently favourable or unfavorable way with respect to a given object
2. Indigenes are natives of a given state or locality, who enjoy some privileges and benefits?
3. Five different constitutions were promulgated for Nigeria between 1900 and 1963
4. The two dominant Air masses in Nigeria are the Tropical Maritime and Continental tropical
5. Nigeria is bounded to the West by The Republic of Benin
6. Nigeria lies between Latitude 4°N and 14°N
7. Nok culture is home to central Nigerian True/False True
8. An advancement in culture is called 27
9. Tourism accrue E for a country (a) Hostility (b) Revenue (c) Dust (d) Carnival
10. Cloth weaving was alien to precolonial Nigerian people YES NO
11. Africa had civilization and cultures dating back to about 4,000 BC TRUE FALSE
12. The last link in channel of distribution is D (a) producer (b) Whole seller (c) Retailer (d) Consumer
13. One of the major difficulties of barter system is B
(a) Double coincidence of wants (b) Presence of common measure of value (c) Divisibility (d) Ease to store and transport
14. Trade relations among countries of the world is called C (a) Unilateral trade (b) Bilateral trade (c) Multilateral trade (d) Regional trade
15. Trade between entities within the political and geo-political boundaries of a country is called C (a) International trade (b) External trade (c) Internal trade (d) Multinational trade

14/20

FEDERAL UNIVERSITY LAFIA
CENTRE FOR GENERAL STUDIES

GST 112 (USE OF LIBRARY, STUDY SKILLS AND ICT)

TEST

Name of student... STANLEY FORTUNE ABENA
Registration Number... 2051600012 Department... MICROBIOLOGY
Faculty... SCIENCES

Instruction: Answer all Questions

1. A Place or building where Information resources are acquired and organized for consultation can be regarded as... Library
2. The act of linking computers so that users can exchange Information or have access to central stored Information is called... Network
3. The physical details of Materials recorded on card catalogue include... Analog and... Audio Visual
4. The direct or indirect services given to library users is called... Reference
5. Academic Library Is a type of Library owned and funded by higher institution of learning.
6. The grouping together of Library materials with common characteristics and separating those that differ in nature is referred to as... cataloging or classification
7. Trade Mark is also called...
8. Library Materials can be divided into... Print and... non-print
9. The Full Meaning of ARPNET is... Advance research project Network
10. Internet can be regarded as an impact of ICT in education.

COURSE CODE: BLY 111. COURSE TITLE: GENERAL BIOLOGY I

TIME ALLOWED: 2.5 HOURS. ANSWER ALL QUESTIONS ON YOUR ANSWER BOOKLET

1. Plants originated from unicellular algae as they colonised the empty landmass.
2. Plants are important because everyday life depends on plants in 5 ways namely, Food, Air, Shelter, medicine and climate.
3. List 5 basic characteristics of plants.
4. Family name used in grouping plants always ends in -ace.
5. List five functions of the genus and specific names of plants.
6. Differentiate briefly between thallophytes, Bryophytes and Pteridophytes.
7. Five (5) characteristics of mosses are small, green, moist, cool and damp area.
8. Name five (5) physical characteristics used in identifying plants: life cycle, form, plant parts, use of local and foliage retention.
9. Three factors that facilitated plants to adapt to life on land include water, land and climate.
10. The Sex Chromosomes X-Y, X-O and Z-W are found in man, Grasshopper and Male (Aves) respectively.
11. Use the principle of monohybrid and dihybrid crosses to give an account of Mendel's first and second laws of heredity.
12. Define evolution and list the evidences of evolution.
13. Cells walls are found in members of these kingdom, except animals, which lack cell walls. a) plants; b) animals; c) bacteria; d) fungi.
14. The nucleolus is the membrane enclosed structure in eukaryotic cells that contains the DNA of the cell. a) mitochondrion; b) chloroplast; c) nucleolus; d) nucleus.
15. The stroma is the fluid-filled space within the chloroplast. a) stroma; b) thylakoid; c) grana; d) stroma.
16. Long, whip-like microfibers that facilitate movement by cells are known as cilia. a) cilia; b) flagella; c) pseudopodia; d) pseudopodia.
17. Ribosomes are constructed in the nucleolus. a) endoplasmic reticulum; b) nucleoid; c) nucleolus; d) nuclear pore.
18. The rough endoplasmic reticulum has its name due to the presence of proteins on its walls. a) proteins; b) amino acids; c) ribosome; d) RNA molecules.
19. Different tissues working together for a specific function is called organ. (a) muscle (b) tendon (c) organ (d) cells
20. Mitosis occurs in All organisms. (a) Humans only (b) Plants only (c) Animals only (d) All organisms
21. Name the structure in the cell that carry out the following function:

(a) controls substances entering or leaving the cell	cell membrane
(b) contains hereditary information	nucleus
(c) releases energy during respiration	mitochondrion
(d) stores and modifies substances by the ER for secretion out of the cell	vacuole golgi body

- ③ ① Autotrophic
- ② multicellular in nature
- ③ have cell wall made of cellulose
- ④ eukaryotic cells
- ⑤ structures adapted for survival.

⑥ - Thallophytes are the simplest plants with no leaves e.g. algae, lichen.
- Bryophytes have no true leaves
- Pteridophytes have leaves and bits of transporting tissues.

⑦ - They are greenish in nature
- have leaves
- Produce their food themselves
- They have rigid bark
- They have roots

FEDERAL UNIVERSITY LAFIA

FACULTY OF SCIENCE

DEPARTMENT OF BOTANY

FIRST SEMESTER EXAMINATIONS 2015/2016 ACADEMIC SESSION BLY 111 (GENERAL BIOLOGY I) PRACTICAL EXAMINATION

INSTRUCTIONS: ANSWER ALL QUESTIONS. TIME ALLOWED: 1 HOUR

- 1) The Safety and emergency response within the laboratory are categorized into -----
-----, -----, ----- and -----
- 2) List 5 Common safety rules in the Laboratory.
i. ----- ii. -----
iii. ----- iv. ----- v. -----
- 3) If the Magnification of an object is 100 with x40 high-dry objective lens. What is the power of the eye-piece? *OCULAR*
- 4) State the use of oil immersion when using X400 lens
- 5) Why is it necessary to stain biological specimen before viewing?
- 6) State two differences between fern and moss plant. *Anterior and Filament*
- 7) Name two reproductive parts of a flower. *Stamen and Ovary*
- 8) What do you observe when onion cell is stained with iodine?
- 9) Would you say the cells are still alive? *Yes*
- 10) List the organelles that are present in onion cell. *Cytoplasm, nucleus, Cell wall, vacuole*
- 11) Highlight two (2) differences between plant and animal cell.
- 12) List the different forms of bacteria. *Cocci, Bacillus, Streptococcus, Spirilla*
- 13) Sketch rhizopus showing the sporangium, sporangiospore, rhizoids, Stolons.
- 14) What is the function of the a Pasteur pipette
- 15) Identify the stages observed after staining the root tips of an onion bulb.
- 16) Apart from acetocarmine what other stain can be used to stain onion root tip. *Papanicolaou*
- 17) Mention five plant that can reproduce sexually. *Unicellular, bryophytes, mosses, Ferns*
- 18) State the function of the following in sexual reproduction
a) Testis b) seminal vesicle c) sperm duct.
- 19) Define the following terms stating an example from each a) Symbiosis b) commensalism
c) parasitism d) competition e) predation.
- 20) What kind of relationship exist between the following
a) Goat and tick - *Parasitism*
b) Remora and shark - *Commensalism*
c) Algae and fungi - *Mutualism*
d) Lion and deer - *Predation*
- 21) List two type of variation in genetics. *Morphological and physiological*
- 22) Give two (2) each of the type of variation listed above
- 23) Calculate the magnification of the following objectives
i. Low power objective
ii. High dry power objective.
iii. Oil immersion.
- 24) State the use of Aceto-carmine in staining. *To enhance visibility to chromosomes*
- 25) Name three types of bacteria with example of each.

Ocular + Obj lens = Mag

FEDERAL UNIVERSITY LAFIA
DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATIONS 2016/2017 SESSION

COURSE TITLE: General Mathematics I

COURSE CODE: MTH 111

INSTRUCTION(S): (1) Do not write on your question paper. (2) Answer all questions.

TIME ALLOWED: $2\frac{1}{2}$ Hours

CREDIT UNIT(S): 3

1. Out of 400 students in Faculty of Science in the Federal University Lafia, 100 took MTH 111 as elective and 150 took PHY 111 as elective. If 75 took both MTH 111 and PHY 111 as electives, how many students took neither MTH 111 nor PHY 111?
(A) 325 (B) 25 (C) 225 (D) 205 (E) 250
2. For what values of x is $2 \cos^2 x - \sin x = 1$ for $x \in [0, 2\pi]$.
(A) $\frac{\pi}{3}, \frac{2\pi}{3}, \pi$ (B) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{4}$
(C) $0, \frac{\pi}{3}, \frac{2\pi}{3}$ (D) $0, \pi, \frac{3\pi}{2}$ (E) $\frac{\pi}{6}, \frac{2\pi}{3}, \frac{3\pi}{2}$
3. If $A = \{1, 2, a, b, c, d\}$, $B = \{2, 3, a, b, c, d, e, f\}$, $C = \{2, 4, c, d, h\}$ and $D = \{1, a, 2, b, 3, c\}$. Obtain $(A \cap B) \cap (C \cup D)$
(A) $\{2, a, b, c, d\}$ (B) $\{1, a, b, 3\}$ (C) $\{2, 1, a, b, c\}$ (D) $\{b, 4, h\}$ (E) $\{1, d\}$
4. Given that $\begin{pmatrix} 3x & 5 \\ -1 & y \end{pmatrix} = \begin{pmatrix} 9 & 5 \\ -1 & -10 \end{pmatrix}$, Find the values of $3x - 2y$?
(A) 29 (B) -28 (C) 13 (D) 10 (E) -27
5. Let $P = \{a, b, c, \dots, h, i\}$, $Q = \{b, d, f, h\}$, $R = \{a, c, e, g, i\}$, $S = \{c, d, e\}$ and $T = \{c, e\}$. If X and Q are disjoint, find X .
(A) R (B) R and T (C) S and T (D) S (E) T
6. Solve the equation $\cos x + \sin x = \frac{1}{\cos x - \sin x}$ for values of x such that $0 \leq x < 2\pi$.
(A) $\frac{\pi}{2}, \frac{3\pi}{2}$ (B) $\frac{\pi}{3}, \frac{2\pi}{3}$ (C) $0, \pi$ (D) $\frac{\pi}{3}, \pi$ (E) $0, \frac{\pi}{3}$
7. The determinant of $P = \begin{pmatrix} \cos 30^\circ & \sin 30^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix}$ is (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{\sqrt{3}}{4}$ (D) 0 (E) $\frac{3}{2}$
8. Express $(3 - 8i)(5 + 7i)$ in the $x + iy$ form. (A) $71 + 24i$ (B) $-41 - 24i$ (C) $-41 + 66i$
(D) $-71 + 66i$ (E) $71 - 24i$
9. If $z = 3 + 5i$, find $z - \bar{z}$. (A) 6 (B) 25 (C) 34 (D) $-4 + 15i$ (E) $10i$
10. One of the following is DeMorgan's law of sets. (A) $(A \cup B)' = (A \cap B)'$ (B) $A \cap A' = \emptyset$
(C) $(A \cup B)' = A' \cap B'$ (D) $(A \cup B)' = A' \cap B'$ (E) $A' \cup A = \xi$
11. If $\cos \theta = \frac{12}{13}$, what is the value of $\sin \theta$? (A) $\frac{13}{5}$ (B) $\frac{6}{13}$ (C) $\frac{5}{13}$ (D) $\frac{13}{6}$ (E) $\frac{5}{12}$
12. The real and imaginary parts of $\frac{7-i}{1+7i}$ are respectively. (A) $\frac{7}{25}, -1$ (B) $0, -1$ (C) $\frac{14}{50}, -\frac{48}{50}$
(D) $\frac{4}{5}, \frac{9}{5}$ (E) $\frac{9}{5}, -\frac{4}{5}$
13. Find two numbers such that their arithmetic mean is 6.5 and their geometric mean is 6.
(A) 4, 9 (B) 4, 16 (C) 6, 8 (D) 3, 6 (E) 5, 7
14. Find the coefficient of x^4 in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$.
(A) ${}^{12}C_8$ (B) ${}^{15}C_3$ (C) ${}^{12}C_5$ (D) 15 (E) 120
15. If the 9th term of an arithmetic sequence is 5 times the 5th term, find the relationship between the first term, a and the common difference, d .
(A) $a + 2d = 0$ (B) $a + 3d = 0$ (C) $2a + 3d = 0$ (D) $2a + d = 0$ (E) $3a + d = 0$

16. Given two matrices S and T with orders (4×5) and (5×6) respectively. The matrix ST has order

- (A) (5×5) (B) (4×5) (C) (4×6) (D) (5×6) (E) (5×4)

17. There are basically _____ step(s) in the proof by mathematical induction. (A) 1 (B) 4 (C) 3 (D) 2 (E) 5

18. If $\theta = \frac{7\pi}{6}$ in radians, what is θ in degrees? (A) 45° (B) 125° (C) 205° (D) 220° (E) 210°

19. In a class of 40 students, 24 play football, 18 play hockey and 6 do not play any game. How many play both? (A) 3 (B) 2 (C) 5 (D) 1 (E) 8

20. What is the n th term of the sequence 9, 3, 1, ...?

- (A) $9\left(\frac{1}{3}\right)^{n+1}$ (B) $9\left(\frac{1}{3}\right)^n$ (C) $27\left(\frac{1}{3}\right)^{n+1}$ (D) $27\left(\frac{1}{3}\right)^n$ (E) $3\left(\frac{1}{3}\right)^{n-1}$

21. The minor of the entry, a_{23} of a given matrix is -14 . What is the value of the cofactor of the entry?

- (A) -14 (B) 13 (C) 14 (D) 15 (E) -13

22. If the sum of infinity of the series $x + x^3 + x^5 + \dots$ is $\frac{2}{3}$, find the value(s) of x . (A) $\frac{1}{2}$ (B) $\frac{1}{2}, -2$

- (C) $\frac{1}{2}, 2$ (D) 2 (E) 3

23. The value of $\tan(-15^\circ)$ in surd form is..... (A) $\sqrt{3} - 2$ (B) $2 - \sqrt{3}$ (C) $\sqrt{3} + 2$

- (D) $2 + \sqrt{3}$ (E) $\sqrt{3} + 3$

24. A diagonal matrix whose diagonal elements are one (1) is called

- (A) Unit matrix (B) Triangular Matrix (C) Diagonal Matrix (D) Null Matrix (E) Square Matrix

25. For what values of r will the equation $x^2 + (r - 1)x + 1 = 0$ have equal roots?

- (A) -1 and -3 (B) 3 and -1 (C) -3 and 1 (D) 1 and 3 (E) 2 and 3

26. If $R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, what is R^{-1} ? (A) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (B) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ (E) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

27. Solve within the range $0^\circ \leq x \leq 180^\circ$ the equation, $\sin x + 2 \sin x \cos x = 0$. (A) $x = 90^\circ, 180^\circ$

- (B) $x = 0^\circ, 150^\circ$ (C) $x = 180^\circ, 360^\circ$ (D) $x = 0^\circ, 120^\circ, 180^\circ$ (E) $x = 0^\circ, 150^\circ, 180^\circ, 360^\circ$

28. If $\cos \theta = \frac{\sqrt{3}}{2}$ and $\tan \theta$ is negative, what is $\sin \theta$? (A) $\frac{\sqrt{2}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $-\frac{\sqrt{3}}{2}$ (E) $-\frac{1}{2}$

29. Find the number of ways of selecting 3 prefects from a class of 18 students.

- (A) 81 (B) 4896 (C) 18 (D) 1253 (E) 816

30. The number, $1 + i$ in polar form is (A) $4(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$ (B) $2(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$

- (C) $\sqrt{2}(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$ (D) $\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$ (E) $\sqrt{2}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

31. Given $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, evaluate $f(A) = A^2 - 4A + 3I$. (A) $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$ (B) $\begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}$

- (C) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ (D) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (E) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

32. Find the determinant of the matrix, $A = \begin{pmatrix} 2 & -6 & 10 \\ 4 & 0 & -2 \\ 3 & -2 & 8 \end{pmatrix}$.

- (A) 0 (B) -316 (C) -26 (D) 208 (E) 120

33. Three consecutive terms in a geometric sequence are $n - 2$, n and $n + 3$. What is the common ratio of the sequence? (A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{3}{2}$ (E) $\frac{4}{3}$

34. Given that $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, find $A^T \cdot A$.

- (A) $\begin{pmatrix} 25 & -2 \\ -2 & 3 \end{pmatrix}$ (B) $\begin{pmatrix} 21 & 2 \\ -2 & 1 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 20 & -2 \\ -4 & 1 \end{pmatrix}$ (E) $\begin{pmatrix} 4 & 2 \\ 0 & -5 \end{pmatrix}$

35. The argument of $-\sqrt{3} + i$ is (A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $-\frac{\pi}{3}$ (E) $\frac{3\pi}{4}$
36. Using the first two terms of series in the binomial expansion of $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} + \dots$.
Evaluate $\sqrt{3}$. (A) $-\frac{1}{2}$ (B) -2 (C) $\frac{7}{4}$ (D) $\frac{1}{4}$ (E) $\frac{1}{2}$
37. The cardinality of the power set of A where $A = \{b, c, d, e\}$ is (A) 4 (B) 10 (C) 8 (D) 16 (E) 32
38. Calculate the argument of $\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi$. (A) π (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$ (E) $\frac{\pi}{4}$
39. Evaluate $\sqrt{-49}$. (A) $12i$ (B) 9 (C) $-7i$ (D) $2\sqrt{2}i$ (E) $7i$
40. If $\theta = 225^\circ$, what is θ in radians? (A) $\frac{5\pi}{6}$ (B) $\frac{4\pi}{5}$ (C) $\frac{6\pi}{5}$ (D) $\frac{5\pi}{4}$ (E) $\frac{4\pi}{3}$
41. The complex number $\sqrt{3} - i$ has modulus as (A) -3 (B) 4 (C) 2 (D) $\frac{\pi}{6}$ (E) $\frac{\pi}{4}$
42. Find the modulus of $\cos \theta + i \sin \theta$. (A) $-i$ (B) 1 (C) -1 (D) i (E) $2i$
43. If α and β are the roots of $x^2 + px + q = 0$ with $\alpha + \beta = 5$ and $\frac{\alpha}{\beta} = \frac{7}{2}$. Find p and q .
(A) $p = -9, q = 14$ (B) $p = 1, q = 10$ (C) $p = -3, q = 10$ (D) $p = 3, q = 0$ (E) $p = 14, q = -9$
44. Find $\tan 150^\circ$. (A) $-\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $-\sqrt{3}$ (E) $\frac{\sqrt{3}}{2}$
45. If $\frac{2+9i}{5-2i} = x + iy$, where x and y are real. What are the values of x and y ?
(A) $\left(-\frac{9}{2}, \frac{2}{5}\right)$ (B) $\left(-\frac{8}{29}, \frac{49}{29}\right)$ (C) $\left(\frac{49}{29}, \frac{8}{29}\right)$ (D) $\left(\frac{2}{5}, -\frac{9}{2}\right)$ (E) $\left(\frac{49}{29}, \frac{8}{29}\right)$
46. A matrix A is skew symmetric if (A) $A^T = -A^T$ (B) $A = A^T$ (C) $A^T = -A$ (D) $A = -A^T$ (E) $A = -A$